

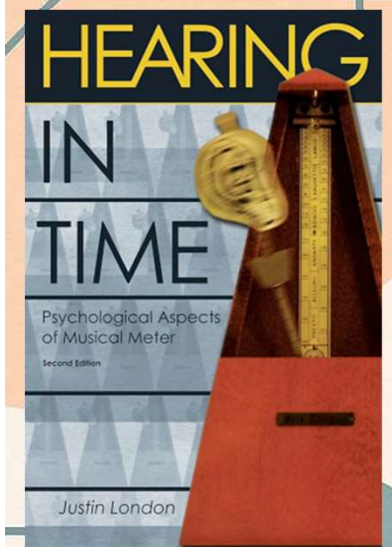
Understanding and Emulating Time: Analyzing and Simulating Musical Microrhythm Timing with the **beat_it** Toolbox

Dr. Egor Polyakov
HMT Leipzig

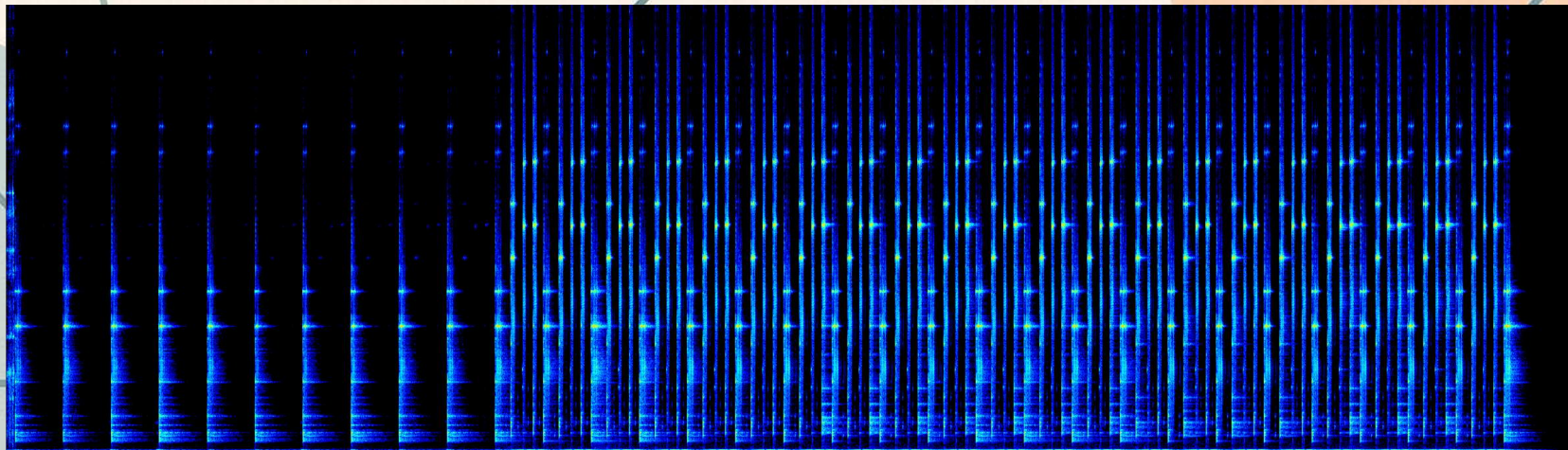
InMusic25 Conference
20-22 June, 2025
Bath Spa University,
Bath, UK

How it all began...

- Microrhythms and microtiming are familiar territory for musicians, but they hadn't been central to my own research or artistic practice—even though some of my drummer and guitarist friends were always passionate about them.
- Even after reading Justin London's book 'Hearing in Time', which deeply influenced me, I still hadn't explored microrhythm personally.
- In 2018, I lectured on the history of electroacoustic music and analyzed several works with students.
- One piece caught our attention: Denis Dufour – 'Bocalises – Prélude' (1977-78).



- Limited information is available about the compositional process, but it is known that the sound material was derived from recordings of various manipulations of glass jars.
- While the impulses clearly align with a metrical grid, how can we analyze the individual microtiming shifts?
- Spectrogram? Waveform? Cuepoints?
- At the time, I primarily worked with IRCAM software such as AudioSculpt, OpenMusic, and Max/MSP — can these tools be used for microtiming analysis?



Spectrogram extracted from Audiosculpt via PNG export

FFT

window size: 2048

window type: hanning

overlap: 8x (12.5%)

oversampling: 4x

Audio

SR: 44100 Hz

Pixel (time resolution)

$2048/8 = 256$ frames

$1/44100s * 2048/8 = 0,0058s$

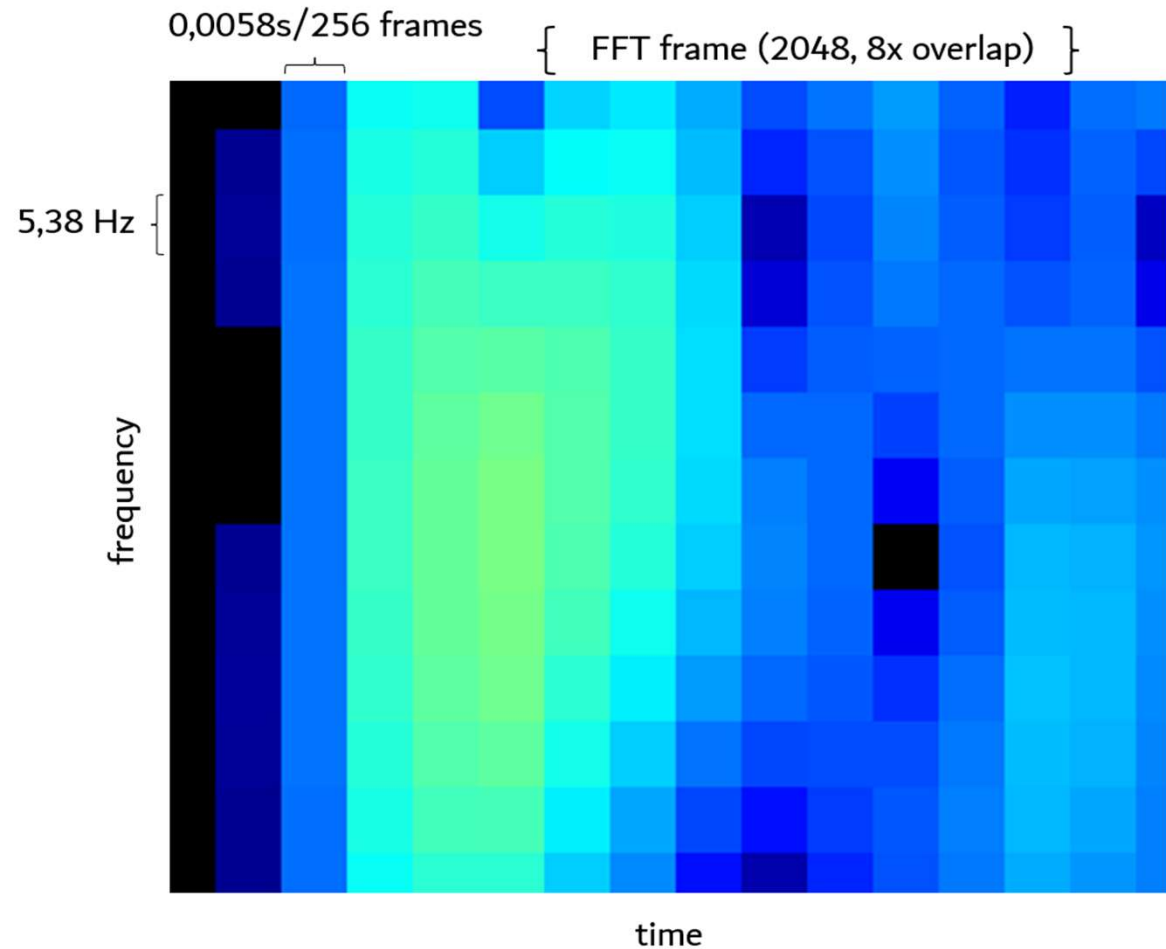
Pixel (frequency resolution)

$FFT\ Size/2 = N\ (Bins)$

$2048/2 * 4 = 4096$ bins

$F_{max}/N(Bins) = FR$

$22050/4096 = 5,38\ Hz$



Calculating time & frequency resolution based on the spectrogram image resolution

Gimp

Reading and modifying spectrogram
via graphic tools.

Use pointer and measure tools to get
exact position/length.

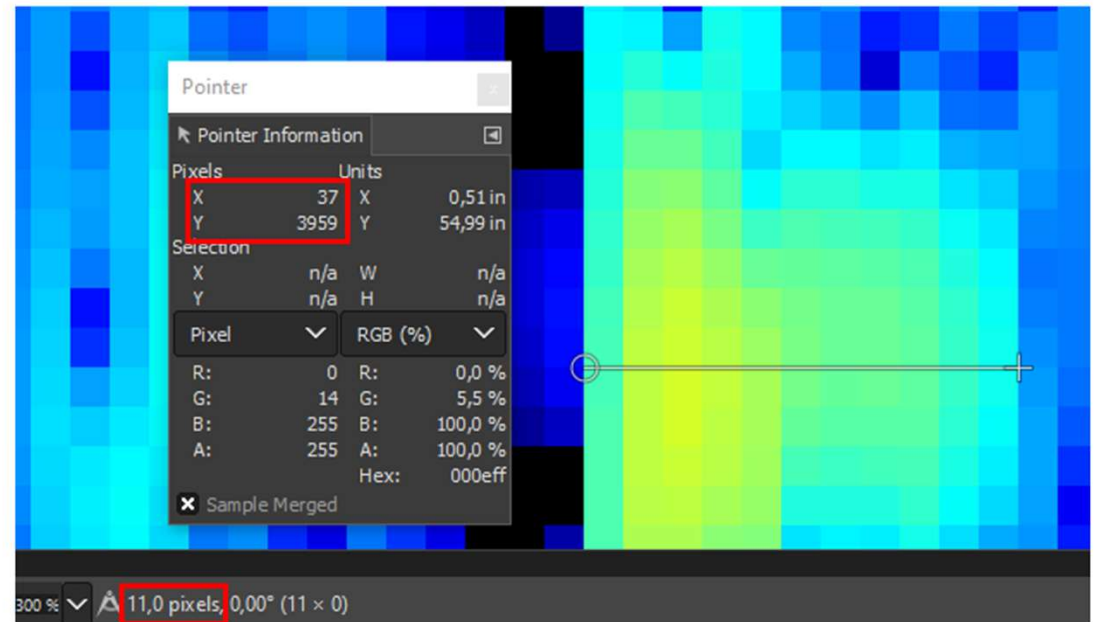
Position:

$$X = 37 = 37 * 0,0058s = 0,2146s$$

$$Y = 3959 = (4096 - 3959) * 5,38 \text{ Hz} = 737,06 \text{ Hz}$$

Length:

$$11 \text{ pixels} = 11 * 0,0058s = 0.0638s$$



Calculating time and frequency values in Gimp [sic] based on spectrogram extracted from Audiosculpt

- AudioSculpt + GIMP hacking was fun, but clearly not a sustainable workflow.
- Core insight that emerged:
*Microtiming = a rhythmic-grid model **plus** a clear visualisation of every impulse's deviation from that grid.*
- Next hurdle: how to mark peaks for analysis?
 - Automatic detection vs. manual annotation?
- **Sonic Visualizer** seemed promising for hand-marking onsets:
 - Manually placing time-instants works, **but** choosing a single reference representation (waveform, spectrogram, onset curve...) is tricky.
 - When you open multiple panes you must keep them in sync yourself (Add Existing Layer → manual alignment), which quickly becomes tedious.
- **Python implementation!**

Whats next?

- **Current Implementation & Accessibility**

- The **beat_it** Toolbox (currently in alpha) is implemented in Python and optimized for Jupyter—excellent for interactive exploration in a web browser, but not straightforward for users without coding experience.
- Cloud platforms like **Google Colab** offer an alternative, yet still require programming skills—and handling your own data isn't always transparent.
- Since 2024, the German NFDI consortium (funded by DFG) offers **Jupyter4NFDI**, a free Jupyter server for students, teachers, and researchers at German universities—hosted by RZ Jülich.
- From September, a pilot program at HfM Weimar, where I hold a senior research position, will onboard tools for symbolic and audio analysis into Jupyter4NFDI.

Thank you!



egor.polyakov@hmt-leipzig.de
<https://github.com/egorpol>

